WEEK 1 PROJECT PROPOSAL

Requirements Analysis and Entity-Relationship Diagram

Class Code: 4-311

1. System Title

Campus Equipment Borrowing and Inventory Management System

2. Problem Description

Universities often lend projectors, cameras, laptops, microphones, and other shared equipment to students and personnel. When these items are monitored using paper logbooks or separate spreadsheets, it becomes difficult to check availability, identify the current borrower, track due dates, and maintain an accurate inventory. This can result in overlapping releases, late returns, misplaced equipment, and incomplete transaction histories.

Students and staff need an organized way to request and borrow equipment, while equipment custodians need reliable information for approval, release, return, and maintenance activities. A relational database system is necessary to connect borrower records, individual equipment items, categories, borrowing transactions, and maintenance reports. These linked records help reduce duplicate information, preserve accountability, and support accurate inventory and borrowing reports.

3. Intended Users

- **Students:** Request available equipment and check their borrowing records.
- **Equipment Staff:** Review requests, issue and receive equipment, and report item condition.
- **Administrators:** Manage user accounts, equipment records, categories, and summary reports.

4. System Objectives

1. Maintain a centralized and accurate inventory of individual equipment units and categories.
2. Record equipment borrowing requests, decisions, releases, and returns.
3. Prevent the same physical equipment unit from being issued to multiple borrowers at once.
4. Document damage and maintenance concerns to help determine which items are safe to lend.
5. Provide searchable transaction histories and basic reports on borrowed, overdue, and available equipment.

5. Scope and Limitations

Scope. The system will store student and staff accounts, equipment categories, uniquely tagged equipment units, borrowing requests, item-level borrowing details, and maintenance records. Equipment staff can approve or reject requests, record releases and returns, and update the condition of an item. Administrators can manage inventory data and obtain basic inventory, active borrowing, and overdue reports.

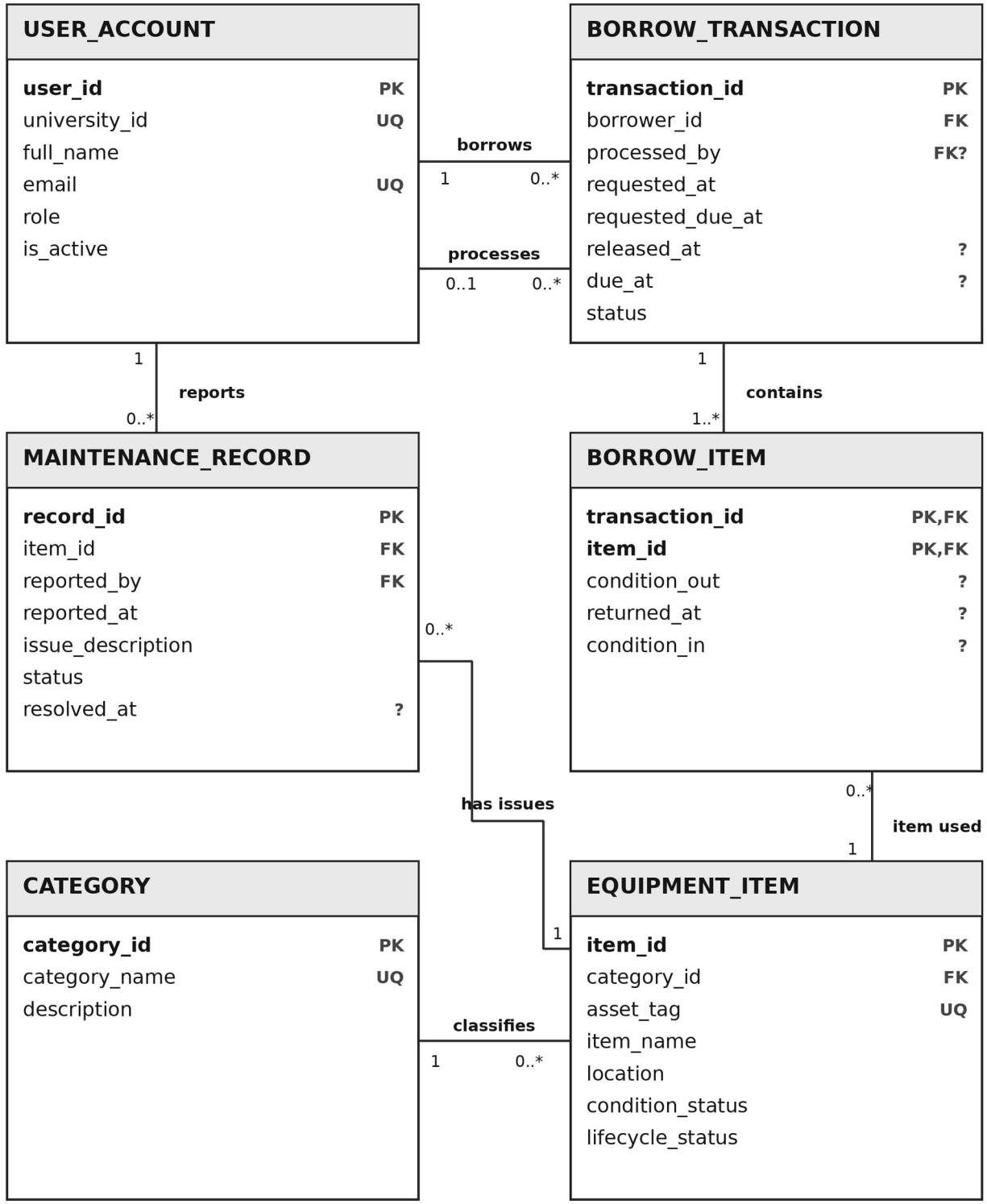
Limitations. The system will not process payments or fines, track equipment using RFID or GPS, send automated SMS or email notifications, or connect to other university systems. It will track reusable, individually identifiable equipment rather than consumable supplies. Availability is finalized when equipment is released, not merely when a request is submitted.

6. Business Rules

1. Each user account has a unique university ID and email address, one role (student, equipment staff, or administrator), and an active or inactive status.
2. An active student or staff account may have zero or many borrowing transactions; every borrowing transaction must have exactly one borrower.
3. Every borrowing transaction must contain at least one borrowing item record.
4. Each borrowing item record belongs to exactly one borrowing transaction and identifies exactly one physical equipment item.
5. An equipment item may appear only once within a particular borrowing transaction.
6. Each equipment item belongs to exactly one equipment category; a category may contain zero or many equipment items.
7. Every physical equipment item has a unique asset tag and may be borrowed many times across different transactions.
8. A physical equipment item cannot be released to more than one borrower at the same time while an earlier release remains unreturned.
9. A request starts with PENDING status and may be approved, rejected, or cancelled; approved requests may proceed to release and later closure.
10. Every approved or rejected request must identify the equipment staff member or administrator who processed it; a pending request may have no processor.
11. Only an approved request may be released, and its due date must be later than its release date and time.
12. The return date/time and return condition of a borrowing item remain blank until that particular equipment unit is returned.
13. A released borrowing transaction may be marked CLOSED only after all of its borrowed equipment items have been returned.
14. Only active equipment in good condition with no unresolved maintenance issue may be released.
15. Every maintenance record refers to exactly one equipment item and exactly one reporting user; an equipment item and a user may each be linked to many maintenance records.

7. Entity-Relationship Diagram (ERD)

The diagram identifies six entities, their attributes, primary and foreign keys, and relationship cardinalities in minimum-maximum notation.



Legend: PK = primary key; FK = foreign key; UQ = unique; ? = nullable. Cardinalities: 1 = exactly one; 0..1 = optional one; 0..* = zero or many; 1..* = one or many.

Participation: Each borrowing transaction must have one borrower and at least one borrowing item. Each borrowing item must reference one transaction and one equipment unit. Each equipment unit must belong to one category. Processing a request is optional until a staff decision is recorded.